

SUMMARY

Dedicated to advancing the fundamentals of machine learning and deep learning. My work bridges theoretical insights with impactful applications in human alignment and machine reasoning, focusing on optimal transport, optimization, and foundation models.

SKILLS

- Technical:** Statistics, Optimization & algorithms, Machine Learning, Deep Learning, LLM.
- Tech-stack:** Python, Java, SQL, Pytorch, FastAPI, AWS Sagemaker, Docker.
- Misc:** Logical & Critical thinking, Leadership, Teamwork, Problem-solving.
- Languages:** Vietnamese (Native), English (Advanced).

EDUCATION

|              |                                                                                                                      |           |
|--------------|----------------------------------------------------------------------------------------------------------------------|-----------|
| 2023 to Now  | <b>Hanoi University of Science and Technology</b><br>Bachelor of Science in Data Science and Artificial Intelligence | 3.91/4.00 |
| 2020 to 2023 | <b>Hanoi National University of Education, High School for Gifted Students</b><br>Math Honors Class                  | 9.6/10    |

RELEVANT EXPERIENCE

|                  |                                                                                                                                               |                                                                  |
|------------------|-----------------------------------------------------------------------------------------------------------------------------------------------|------------------------------------------------------------------|
| 2026.8 to Now    | <b>AI Research Resident</b>                                                                                                                   | Qualcomm AI Research                                             |
|                  | • Onboard soon                                                                                                                                |                                                                  |
| 2025.4 to 2025.8 | <b>Internship - Viettel Digital Talent</b>                                                                                                    | Viettel                                                          |
|                  | • Prototyped reasoning-focused LLM for networking application, improved accuracy by 4% on internal test dataset.                              |                                                                  |
| 2024.6 to Now    | <b>Research Assistant</b>                                                                                                                     | The International Research Center for AI – Foundation Models Lab |
|                  | • My research focuses on human alignment, machine reasoning, and the fundamentals of deep learning under the supervision of Dr. Ngo Van Linh. |                                                                  |

PUBLICATIONS

For an updated list, please refer to my [Google Scholar](#) profile.

|      |                                                                                                                       |                      |
|------|-----------------------------------------------------------------------------------------------------------------------|----------------------|
| 2026 | <b>BSO: Safety Alignment Is Density Ratio Matching</b>                                                                | <a href="#">Link</a> |
|      | Under review.                                                                                                         |                      |
| 2026 | <b>Spectral Flattening Is All Muon Needs: How Orthogonalization Controls Learning Rate and Convergence</b>            | <a href="#">Link</a> |
|      | Under review.                                                                                                         |                      |
| 2026 | <b>Route-and-Generate: KL-Regularized Router Improvement for Reasoning Routing in LLMs</b>                            | <a href="#">Link</a> |
|      | Under review.                                                                                                         |                      |
| 2026 | <b>Principled Token-Level Preference Optimization via Ratio Matching</b>                                              | <a href="#">Link</a> |
|      | ICML 2026 (First Author).                                                                                             |                      |
| 2025 | <b>CTPD: Cross Tokenizer Preference Distillation</b>                                                                  | <a href="#">Link</a> |
|      | AAAI 2026 (First Author).                                                                                             |                      |
| 2025 | <b>FAID: Fine-grained AI-generated Text Detection using Multi-task Auxiliary and Multi-level Contrastive Learning</b> | <a href="#">Link</a> |
|      | EACL 2026 (Co-author).                                                                                                |                      |

HONORS AND AWARDS

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|------|------------------------------------------------------------------------------------------------------------------------------------------------|--|
| 2025 | HUST Academic Excellence Award.                                                                                                                |  |
| 2025 | Second Prize - Student Scientific Research Competition.<br><b>(Machine-generated text detection system (multilingual, multi-domain))</b>       |  |
| 2024 | HUST Academic Encouragement Scholarship 2024.1 Semester. <b>(Dean's List-equivalent merit scholarship)</b>                                     |  |
| 2024 | Top 6 (5%) SOICT Hackathon AI Application track. <b>(Automatic data visualization platform, improving data analysis speed and performance)</b> |  |
| 2023 | HUST Academic Encouragement Scholarship 2023.1 Semester. <b>(Dean's List-equivalent merit scholarship)</b>                                     |  |
| 2023 | Second Prize in Mathematics, HNUE Olympiad 2023.                                                                                               |  |
| 2022 | Third Prize in Mathematics, HNUE Olympiad 2022.                                                                                                |  |

## CERTIFICATIONS

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| 2022 | 7.0 IELTS. |
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## EXTRACURRICULAR EXPERIENCES

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| 2023.9 to Now | <b>Member of Study, Scientific Research &amp; Orientation</b>                                                                                                                                                                                                                                    | <b>HCM Youth Union</b>       |
|               | <ul style="list-style-type: none"> <li>Led student-driven initiatives to promote research activities, organizing events and mentorship programs to improve academic engagement.</li> <li>Assisted students in scientific research, career development, and skill-building activities.</li> </ul> |                              |
| 2025.6 to Now | <b>Head of Technical Committee</b>                                                                                                                                                                                                                                                               | <b>SoICT Innovation Club</b> |
|               | <ul style="list-style-type: none"> <li>Organized seminars and workshops; published slides and relevant documents.</li> <li>Mentored new members through a 6-week track; shipped multiple capstones.</li> </ul>                                                                                   |                              |

## REFERENCES

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### Dr. Ngo Van Linh

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### Dr. Trung Le

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